

DGP-by-Estimator Monte Carlo

Methods, corrected results, and interpretation

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Abstract

This simulation applies four procedures—Kline–Saggio–Sølvsten (KSS), Bonhomme–Lamadon–Manresa (BLM), Borovičková–Shimer (BS20), and the current project low-rank plug-in—to panels generated from five distinct data-generating processes (DGPs). A DGP is the rule that creates the full worker–firm wage schedule, the assignment probabilities, and the sampled panel. The project leave-one-out (LOO) correction is still under development and is not simulated. The central result is that three distinct issues must be kept separate: sampling error, disagreement between population estimands, and whether an estimator’s required support is present. A forensic audit showed that the original BLM summaries mixed supported fits with fits missing stayer-class support. The corrected pipeline rejects such samples before estimation. Of 500 BLM attempts, 243 have complete support, 242 pass the likelihood diagnostic, one returns with a numerical warning, and 257 are classified as unsupported. No supported BLM attempt fails in PyTwoWay. All results below use this corrected classification.

1 Question and scope

The experiment asks one symmetric question: what happens when every procedure is used under every DGP? This is more informative than generating all panels from an additive Abowd–Kramarz–Margolis (AKM) model because an additive design mechanically favors an additive estimator. The five DGPs are additive AKM, Crippa/Tukey nonadditivity, discrete BLM types, continuous low-rank factors, and Gibbons–Katz–Lemieux–Parent (GKLP) comparative advantage. Each DGP has 100 independent Monte Carlo replications. One Monte Carlo replication means drawing a new finite population, sampling one panel from it, running every procedure, and comparing each returned number with that population’s known truth.

The project procedure in this document is the current Bayesian information criterion (BIC)-selected low-rank plug-in. It is not called the final project LOO estimator because the full LOO correction has not yet been implemented.

The argument proceeds in the order needed to interpret a simulation result. Section 2 defines the four economic quantities to be recovered. Section 3 states every DGP parameter. Section 4 explains what each procedure actually returns and which project quantities, if any, can be constructed from it. Section 5 defines feasibility, bias, and root mean squared error. The results then ask, in order: can the procedure be run, does it work in its own benchmark, how accurate is it for a common target, are native and project targets different, does rank selection recover the interaction dimension, and do warning fits create tail risk. This order prevents a low error computed on a selected subset from being mistaken for overall good performance.

Every reported number is calculated from the support-audited merged output. Its configuration fingerprint is 3e8ab3b57d35190989c47117f139b414b9f656db8a933d6c5d1df6d2aa5fbcc4. The audit re-constructs panels and estimated BLM firm classes from the saved seeds, but does not refit a successful estimator. Statements borrowed from a paper or software specification are cited where used; otherwise, equations are definitions or explicit derivations.

2 Notation and project estimands

Table 1 gives every symbol needed later. Starting here avoids changing notation as the procedures are introduced.

Table 1: Notation used throughout the experiment

Symbol	Exact definition	Meaning
i, j, t	Worker, firm, and period indices	Unit of the matched panel
Y_{ijt}	$m_{ij} + \varepsilon_{ijt}$	Observed wage outcome
m_{ij}	$\mathbb{E}[Y_{ijt} \mid i, j]$	Complete systematic wage schedule
P_{ij}	Population probability of match (i, j)	Observed assignment law; $\sum_{ij} P_{ij} = 1$
p_i, q_j	Marginals of P_{ij}	Worker and firm population weights
$p_i q_j$	Product of the two marginals	Independent-assignment counterfactual
$\mathbb{E}_P, \mathbb{E}_{pq}$	Expectations weighted by P_{ij} and $p_i q_j$	Observed assignment and independent assignment
a_i, b_j, h_{ij}	Components of $m_{ij} = \mu + a_i + b_j + h_{ij}$ under $p_i q_j$	Worker main effect, firm main effect, and interaction
Q_F	$\mathbb{E}_{pq}[(m_{ij} - \mathbb{E}_q[m_{ij} \mid i])^2]$	Firm-side schedule variation
H_F	$2\mathbb{E}_{pq}[h_{ij}^2]$	Twice the nonadditive interaction variance
ρ_H	$H_F / (2Q_F)$ when $Q_F > 0$	Interaction share of firm-side variation
C_{assign}	$\{\text{Var}_P(m) - \text{Var}_{pq}(m)\} / 2$	Contribution of observed assignment to schedule variance
α_i, ψ_j	Pseudo-true additive worker and firm effects	Native KSS/AKM projection objects
λ_i, μ_j	$\mathbb{E}_P[m_{ij} \mid i]$ and $\mathbb{E}_P[m_{ij} \mid j]$	Native BS20 worker and firm wage types

The decomposition is unique after defining

$$\mu = \mathbb{E}_{pq}[m_{ij}], \quad a_i = \mathbb{E}_q[m_{ij} \mid i] - \mu, \quad (1)$$

$$b_j = \mathbb{E}_p[m_{ij} \mid j] - \mu, \quad h_{ij} = m_{ij} - \mu - a_i - b_j. \quad (2)$$

Thus the p -weighted mean of a_i , the q -weighted mean of b_j , and both weighted margins of h_{ij} are zero. Consequently,

$$m_{ij} - \mathbb{E}_q[m_{ij} \mid i] = b_j + h_{ij}, \quad (3)$$

$$Q_F = \text{Var}_q(b_j) + \mathbb{E}_{pq}[h_{ij}^2] = \text{Var}_q(b_j) + \frac{H_F}{2}. \quad (4)$$

Equation (4) is a direct derivation from the project definitions, not an assumption about an estimator.

2.1 Why these four project quantities are reported

The four quantities answer different economic questions and are therefore not interchangeable. Q_F asks how much the systematic wage schedule changes on the firm side after holding the worker fixed. H_F asks how much of the schedule is genuinely nonadditive, meaning that firm differences depend on the worker. The factor 2 in H_F is chosen so that the accounting identity in Equation (4) has the simple term $H_F/2$. ρ_H divides the interaction contribution $H_F/2$ by total firm-side variation Q_F ; it is a unit-free interaction share and is easier to compare across DGPs of different scale. Finally, C_{assign} asks how much observed sorting changes schedule variance relative to random matching with the same worker and firm marginals. The factor 1/2 makes $C_{\text{assign}} = \text{Cov}_P(a_i, b_j)$ in the additive case.

Together the four objects distinguish the amount of firm-side heterogeneity (Q_F), its nonadditive part (H_F and ρ_H), and the assignment channel (C_{assign}). Worker-side analogues are omitted because the project question in this experiment is explicitly firm-side. Q_F , H_F , and C_{assign} are measured in squared wage units; ρ_H is unitless.

3 Data-generating processes

3.1 Common population and panel design

Every population contains 300 workers and 18 firms. Every panel contains ten periods, hence 3,000 observations before procedure-specific cleaning. The systematic wage schedule has grand mean zero. Worker and firm population marginals are uniform. The observation disturbance is independent and normally distributed with mean zero and standard deviation 0.5.

Because this is a simulation, the complete 300×18 schedule and all P_{ij} are known before the panel is sampled. The population truths in Section 2 are therefore computed by exact finite sums over all 5,400 worker–firm cells. They are not estimated from the simulated panel. The base random seed is 20260804, so every population and panel is reproducible.

All Gaussian coordinates are redrawn in every replication. Main-effect vectors are standardized to finite-population mean zero and variance one. Factor columns are centered and orthonormalized under the uniform population weights: each column has weighted mean zero and weighted variance one, and distinct columns have weighted covariance zero. A singular value below is the strength of one worker–factor–firm–factor interaction direction. The stated singular values therefore hold exactly in every realized population, rather than only in expectation.

The rank of an interaction is the number of nonzero worker–factor–firm–factor directions: rank zero is additive, rank one has one such direction, and so on. In Table 2, U_i and V_j denote row vectors of normalized factor scores, and a prime denotes transposition. $N(0, 1)$ denotes a normal draw with mean zero and variance one before the finite-population normalization.

Observed matching probabilities are defined by

$$P_{ij} = R_i S_j \exp \left\{ 0.5 a_i b_j + 0.3 \frac{h_{ij}}{\sqrt{\mathbb{E}_{pq}[h_{ij}^2]}} \right\}, \quad (5)$$

where a_i, b_j, h_{ij} are the realized components of the relevant DGP and the positive balancing constants R_i and S_j restore uniform worker and firm marginals. For AKM, $h_{ij} = 0$ and the second sorting term is omitted. Thus the coefficient 0.5 controls sorting on common worker and firm components, while 0.3 controls sorting on match-specific gains. Define a worker’s conditional firm probabilities by $\pi_{ij} = P_{ij}/p_i$.

The first firm is drawn from π_i . In each later period, the worker makes an independent draw from π_i with probability 0.40 and otherwise remains at the current firm. Because the new draw can equal the current firm, a redraw need not create an observed move.

3.2 The five wage schedules

Table 2: DGP-specific structure; all omitted common parameters are stated above

DGP	Construction	Nonadditivity
AKM	Independent raw worker and firm $N(0, 1)$ effects	$m_{ij} = \alpha_i + \psi_j$; rank zero [1]
Crippa/Tukey	Independent raw $\alpha_i, \psi_j \sim N(0, 1)$	$m_{ij} = \alpha_i + \psi_j + 0.75\alpha_i\psi_j$; rank one with singular value 0.75 [4, Sec. 2.2]
BLM types	Workers are randomly divided equally into types $G_i \in \{1, 2\}$ and firms into classes $K_j \in \{1, 2, 3\}$; raw type main effects and factor entries are $N(0, 1)$	$m_{ij} = A_{G_i} + B_{K_j} + U_{G_i}V_{K_j}$ after normalization; rank one with singular value 1 [2]
Low-rank factors	Raw worker factor coordinates have correlation 0.35 with α_i ; raw firm coordinates have correlation 0.25 with ψ_j	$m_{ij} = \alpha_i + \psi_j + U_i' \text{diag}(1, 0.5)V_j$; rank two
GKLP	Raw $\text{Corr}(Z_i, \eta_i) = 0.35$ and $\text{Corr}(c_j, b_j) = 0.25$	$m_{ij} = Z_i + c_j + b_j\eta_i + \frac{1}{2}b_j^2(0.5)^2$, recentered to mean zero; rank one [5, 6]

The Crippa and GKLP rows adapt the cited wage surfaces to the finite simulated population. The numerical standardization, assignment law in Equation (5), and parameter values in Table 2 are simulation choices, not claims from those papers.

4 Procedures

4.1 KSS

KSS applies leave-out variance-component corrections to a linear model with many effects and unrestricted heteroskedasticity [7]. Here PyTwoWay 0.3.21 supplies the heteroskedastic KSS correction with approximate trace and leverage calculations (`fe_exact=False`) [8]. Its native population target is the best assignment-weighted additive approximation:

$$(\mu, \alpha, \psi) = \arg \min_{\mu, \alpha, \psi} \mathbb{E}_P[(m_{ij} - \mu - \alpha_i - \psi_j)^2]. \quad (6)$$

This target equals the true schedule under AKM. Under a nonadditive DGP it is “pseudo-true”: it is the closest additive schedule, not the complete schedule. KSS natively reports $\text{Var}_q(\psi_j)$ and $\text{Cov}_P(\alpha_i, \psi_j)$.

The code calculates three versions. AKM-FE is the uncorrected fixed-effect plug-in. KSS-HO applies the homoskedastic leave-out correction, which assumes one error variance. KSS-HE allows observation-specific error variances and is the main KSS specification graphed below. Showing one corrected KSS column keeps the DGP-by-procedure comparison readable; KSS-HE is chosen because it maintains the weaker error-variance assumption. The AKM-FE and KSS-HO outputs remain in the machine-readable results.

Before estimation, PyTwoWay keeps a leave-one-spell-connected sample: the worker–firm network remains connected after any single continuous employment spell is removed. It collapses observations when computing that network, drops workers observed at only one firm for only one spell, and retains return spells. This cleaning is procedure-specific; the DGP itself is unchanged.

The additive restriction also implies

$$h_{ij}^{\text{KSS}} = 0, \quad H_F^{\text{KSS}} = 0, \quad \rho_H^{\text{KSS}} = 0 \quad \text{if } Q_F^{\text{KSS}} > 0. \quad (7)$$

These are imposed zeros, not interaction quantities estimated from the data. The corrected report includes them to show exactly what additivity implies. KSS residual variance is not relabeled as H_F : under misspecification, that residual mixes observation noise with omitted schedule structure.

4.2 BLM

BLM models a discrete worker type by firm class wage distribution and estimates firm classes before the finite-mixture wage model [2]. A wage CDF is the cumulative distribution function of wages; its value at a threshold is the share of observations at or below that threshold. Every DGP is fit using two worker types, three estimated firm classes, ten CDF evaluation points, four random initializations, the best two retained starts, at most 250 mover and stayer iterations, and convergence threshold 10^{-6} . Multiple starts are used because a finite-mixture likelihood can have several local optima. The implementation is PyTwoWay 0.3.21 [8].

The actual pipeline is linear:

1. Drop every worker who leaves a firm and later returns to it.
2. Collapse the remaining panel to employment spells.
3. Estimate three firm clusters from wage CDF features using K-means.
4. Convert the spell panel to mover and stayer event-study observations.
5. Require all three stayer firm classes and all nine mover origin–destination class pairs.
6. Fit four mover-model starts and retain the two best likelihoods.
7. Among the two highest-likelihood starts, select the one with the highest minimum mover-graph connectedness across worker types.
8. Fit the stayer model conditional on the selected mover model.

An employment spell is a worker’s uninterrupted run at one firm. A mover event joins the worker’s origin and destination spells; a stayer event records a worker observed at one firm throughout. K-means is the clustering algorithm that groups firms with similar wage-CDF features.

Step 5 is the audit correction. PyTwoWay 0.3.21 reshapes probability counts to 2×3^2 for movers and 2×3 for stayers without padding absent categories. Without a preflight support check, a missing terminal label raises a reshape exception, while a missing interior label can instead create a zero probability row. The corrected code detects the common economic cause before calling the optimizer.

BLM’s worker types and firm classes are latent, so their numerical labels have no intrinsic order. For simulation scoring only, the estimated 2×3 cell-mean table is relabeled by the worker- and firm-label permutation that minimizes cell-mean squared error relative to a known evaluation table. Under the

grouped BLM DGP, that evaluation table uses the simulated true types. For the other DGPs, workers and firms are placed into deterministic equal-count bins ordered by their product-marginal mean wages. These truth-based (“oracle”) labels are never given to the BLM estimator.

The four project functionals are then applied to the aligned estimated cell means using the known population probability mass of each evaluation cell. This use of population cell weights is another simulation-only scoring choice: it isolates error in BLM’s wage cell means rather than also asking BLM to estimate the assignment law. Finally, those grouped functionals are compared with the functionals of the full 300×18 project schedule. Therefore the reported BLM error outside its discrete DGP includes the deliberate loss from compressing a continuous schedule into six cells.

4.3 BS20

BS20 estimates moments of worker and firm wage types in matched data [3]. PyTwoWay’s weighted standard estimator is used after strongly-connected, spell-level, no-return cleaning [8]. The native population covariance is

$$C_{BS} = \text{Cov}_P(\lambda_i, \mu_j), \quad \lambda_i = \mathbb{E}_P[m_{ij} \mid i], \quad \mu_j = \mathbb{E}_P[m_{ij} \mid j]. \quad (8)$$

The code reads PyTwoWay’s `cov(lambda, mu)` directly. It does not construct that number from project quantities. Because C_{BS} is not generally C_{assign} , BS20 is no longer displayed as an estimator of C_{assign} . Their population difference is reported separately as an estimand comparison.

Concretely, “strongly connected” means every retained firm can be reached from every other through directed worker moves. The panel is collapsed to one row per continuous employment spell, return spells are removed, and each retained worker and firm must appear in at least two spell rows. These choices follow the configured PyTwoWay BS20 pipeline and are not imposed on the other procedures.

4.4 Project low-rank plug-in

The current project fit is

$$m_{ij} = \alpha_i + \psi_j + \sum_{\ell=1}^r s_{\ell} U_{i\ell} V_{j\ell}. \quad (9)$$

Here $U_{i\ell}$ and $V_{j\ell}$ are normalized worker and firm factor scores, s_{ℓ} is the strength of factor ℓ , and r is the number of retained interaction directions. Positive ranks use three starts, tolerance 10^{-6} , and at most 300 iterations. Candidate ranks are $\{0, 1\}$ for AKM and grouped BLM and $\{0, 1, 2\}$ for Crippa, continuous low-rank, and GKLP. All candidates use a common retained core: minimum degree 3 for $\{0, 1\}$ and degree 4 for $\{0, 1, 2\}$.

The selected rank minimizes the Bayesian information criterion (BIC)

$$\text{BIC}(r) = n \log(\text{SSE}_r / n) + \text{df}(r) \log n, \quad \text{df}(r) = N + J - 1 + r(N + J - 2 - r). \quad (10)$$

Equation (10) is an exploratory simulation rule. It is not part of the unfinished project LOO theory. Here SSE_r is the weighted sum of squared observed match-mean residuals at rank r , n is the retained number of wage observations, and N and J are the retained numbers of workers and firms. The first term rewards fit; the degrees-of-freedom penalty discourages extra interaction dimensions.

The plug-in first retains a connected worker–firm core with the stated minimum degree, meaning that every retained worker and firm has at least three or four distinct observed partners as specified above.

It estimates observed match means and completes the unobserved entries of the schedule using Equation (9). It then applies the definitions in Section 2 to the completed schedule and the empirical assignment weights in that retained core. Comparing those estimates with the full-population truth therefore includes sampling, core selection, completion, and plug-in error.

4.5 What can be compared with a common project target

Table 3 is the translation used in every later table and graph. “Cross-target” means that the procedure’s native output is compared with a different project quantity; this is informative about misspecification but is not native-estimand bias. “Imposed” means the value follows from a model restriction and was not estimated. A dash means that no defensible translation was made.

Table 3: Procedure outputs and their mapping to the four project quantities

Procedure	Q_F column	H_F column	ρ_H column	C_{assign} column
KSS-HE	$\text{Var}_q(\psi_j)$, cross-target	0, imposed	0, imposed	$\text{Cov}_P(\alpha_i, \psi_j)$, cross-target
BLM	Grouped cell schedule	Grouped cell schedule	Grouped cell schedule	Grouped cell schedule
BS20	–	–	–	–; its native output is C_{BS}
Project plug-in	Completed schedule	Completed schedule	Completed schedule	Completed schedule

5 Evaluation rules

Each procedure is run once in each of 100 independent replications of each DGP. For a returned estimate $\hat{\theta}_s$ and its replication-specific population truth θ_s , let S be the number of admissible returned estimates in the row being summarized and define

$$e_s = \hat{\theta}_s - \theta_s, \tag{11}$$

$$\text{Bias} = \frac{1}{S} \sum_{s=1}^S e_s, \quad \text{RMSE} = \sqrt{\frac{1}{S} \sum_{s=1}^S e_s^2}. \tag{12}$$

Bias answers whether errors point systematically upward or downward. RMSE combines systematic error and dispersion in one number and penalizes rare large errors, so it is the primary accuracy metric in the cross-procedure graph. Because the four economic quantities have different scales, RMSE values should be compared across procedures within a metric panel, not across different metric panels. Bias is reported where its sign is economically informative, rather than adding a second large heatmap that would obscure the main comparison.

Before calculating either accuracy measure, the experiment records whether a procedure could legitimately return an estimate. This is necessary because a small RMSE on a selected minority of panels does not imply that a procedure is generally feasible. The experiment separates four statuses:

- **Pass/completed.** KSS and BS20 returned a value. BLM additionally has complete support and no one-step likelihood decline greater than 10^{-4} . The project fit satisfies convergence and multi-start diagnostics.

- **Returned with warning.** An admissible BLM or project fit returned finite values but failed its procedure-specific numerical diagnostic.
- **Unsupported.** BLM lacks a stayer class or mover class-pair after its own cleaning and clustering. It is not sent to the optimizer.
- **Estimator failure.** An admissible sample reaches the estimator, but the estimator returns no value.

The 10^{-4} BLM likelihood threshold is an engineering diagnostic, not a statistical critical value. It is 100 times the 10^{-6} optimizer tolerance: the wider band ignores rounding-scale movement while flagging a meaningful decrease in a likelihood that should improve during the update. For a positive-rank project fit, at least two of three starts must have weighted squared-error objectives within $0.0001 \max(1, O^*)$ of the best objective O^* , and their Q_F , H_F , and C_{assign} spreads must each be within $0.001 \max(1, |\theta^*|)$, where θ^* is the corresponding quantity from the best start. These thresholds are also engineering checks. A second near-best start is required so that the solution is not supported by a single favorable initialization. Scaling by $\max(1, |\theta^*|)$ combines an absolute tolerance near zero with a relative tolerance for large quantities; 0.001 means agreement within one thousandth on that scale. A pass does not prove correct specification, correct rank, or low bias.

RMSE uses all values returned from admissible samples. Unsupported samples do not enter the RMSE because no estimate exists. Here “admissible returned” means pass plus returned-with-warning. A pass-only RMSE is reported separately as a numerical diagnostic and is not treated as unconditional procedure performance.

Table 4 states why each subsequent display is included. The exact-count tables and graphical summaries are intentionally paired: the tables support precise lookup, while the graphs make patterns across the full DGP-by-procedure matrix visible at a glance.

Table 4: Question answered by each results display

Display	Question	Why it is needed
Status table and mosaic	Can the procedure be run, and did its diagnostic pass?	Prevents conditional accuracy from hiding lack of support or numerical warnings
Own-model benchmark table	Does an established procedure recover a target when its structure matches the DGP?	Checks the implementation before interpreting misspecification experiments
Common-target RMSE heatmap	Which mapped procedure is closest to the same project truth?	Supplies the main apples-to-apples accuracy comparison
Native-minus-project gaps	Are procedures estimating different population objects before sampling begins?	Separates estimand disagreement from estimation error
Rank table and heatmap	Does BIC retain the true number of interaction directions?	Distinguishes model-selection error from numerical convergence and plug-in bias
All-returned versus pass-only RMSE	Do warning fits create large squared-error tails?	Shows whether average performance is driven by a small unstable subset

6 Results

6.1 Feasibility and numerical status

Table 5 and Figure 1 report pass/warning/unsupported/failure counts. For KSS and BS20, “pass” means only that PyTwoWay completed. The table gives exact counts; the mosaic turns each count into a proportional colored segment so differences across 20 DGP-procedure cells are easier to see. Neither display is an accuracy comparison.

Table 5: Status counts out of 100 replications: pass / warning / unsupported / failure

DGP	KSS-HE	BLM	BS20	Project
AKM	100/0/0/0	43/0/57/0	100/0/0/0	100/0/0/0
Crippa/Tukey	100/0/0/0	57/0/43/0	100/0/0/0	100/0/0/0
BLM types	100/0/0/0	46/1/53/0	100/0/0/0	6/94/0/0
Low-rank factors	100/0/0/0	46/0/54/0	100/0/0/0	78/22/0/0
GKLP	100/0/0/0	50/0/50/0	100/0/0/0	94/6/0/0

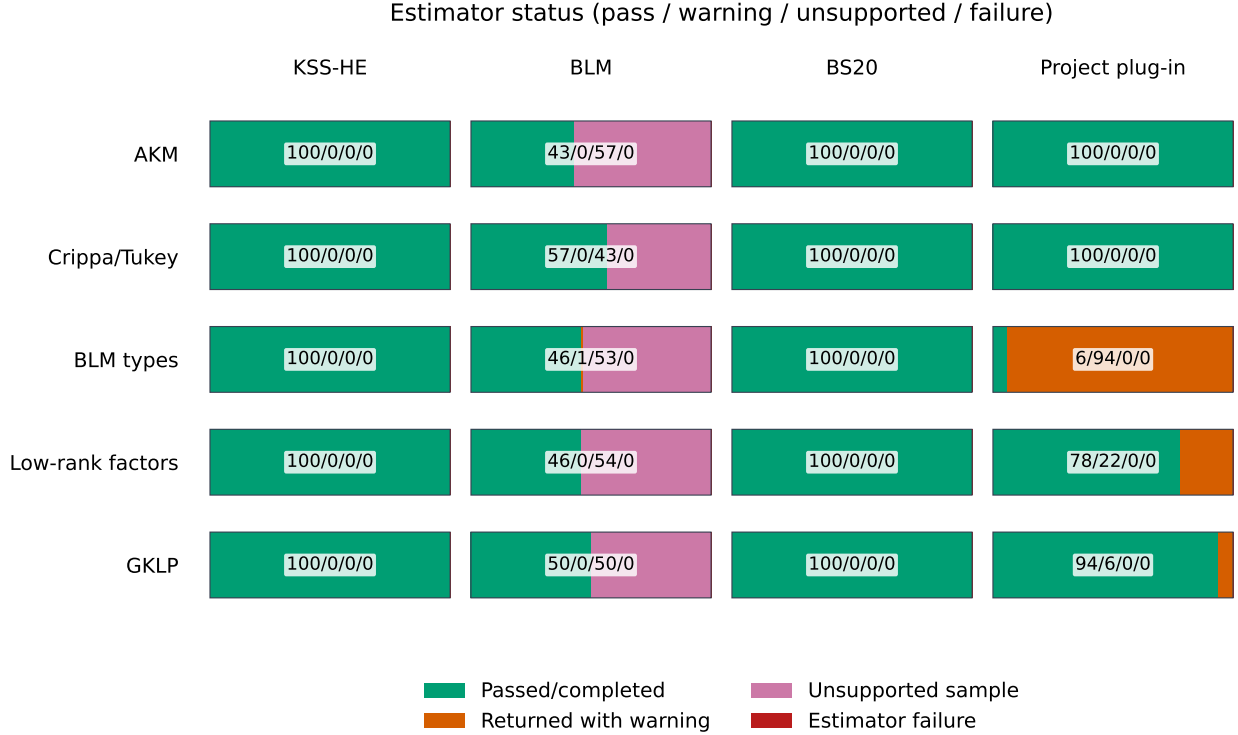


Figure 1: Feasibility and numerical status across all 100 replications per cell. Segment widths are proportional to their counts. The four printed numbers are pass / warning / unsupported / failure, as defined in Section 5.

The BLM audit finds 257 unsupported samples and zero estimator failures. The unsupported samples consist of 3 with no stayers, 246 with at least one absent stayer class, 6 with only a missing mover class-pair, and 2 with both a missing stayer class and mover class-pair. Thus 243 samples have complete support; 242 pass and one returns with a likelihood-path warning.

The high unsupported rate follows from the panel design rather than random-start failure. To see the mechanism, consider the uniform-choice benchmark in which a worker chooses each of 18 firms with probability 1/18. The probability of no actual firm change in a later period is then

$$0.6 + \frac{0.4}{18} = 0.622. \quad (13)$$

The probability of remaining at the same firm through nine later periods is $0.622^9 = 0.014$, or about four workers out of 300. Sorting makes firm choices worker-specific, so this is an explanatory benchmark rather than the exact probability in Equation (5). After returners are dropped, the realized stayers must cover all three firm classes. The reported classification always uses the exact cleaned-panel support, not this benchmark.

6.2 Correctly specified benchmarks

This subsection is an implementation check rather than the final horse race. It focuses on the two cleanest established-procedure benchmarks in the DGP ladder: additive KSS under AKM and discrete BLM under the grouped DGP. BS20 has no corresponding BS20 DGP in the ladder, and the unfinished project procedure’s rank behavior is examined separately. Under AKM, KSS-HE’s bias for Q_F is +0.0032 and its RMSE is 0.0485; its bias for C_{assign} is −0.00014 and its RMSE is 0.0314. Two KSS moments fit in the sentence, whereas BLM maps to all four project functionals and therefore receives Table 6. Under the grouped BLM DGP, the 47 admissible returned BLM fits have the results in Table 6. These are conditional on support and therefore do not erase the 53 percent infeasibility rate.

Table 6: BLM accuracy for the four project functionals under its grouped DGP, among 47 admissible returned fits (46 passes and one warning). Bias and RMSE are defined in Equation (12).

Project functional	Bias	RMSE
Q_F	−0.0054	0.0492
H_F	−0.0031	0.0741
ρ_H	+0.00056	0.0140
C_{assign}	+0.0025	0.0110

6.3 Common project targets

Figure 2 compares admissible returned outputs with the common project truth. This is the central performance display because every filled cell uses the same replication-specific target named by its panel. KSS-HE, rather than AKM-FE or KSS-HO, is shown for the reason given in Section 4.1. KSS-HE is shown for H_F and ρ_H because Equation (7) implies structural zeros. The comparison therefore measures what the additive restriction misses; it is not an estimated KSS interaction. BS20 is absent from the C_{assign} panel because Equation (8) defines a different covariance.

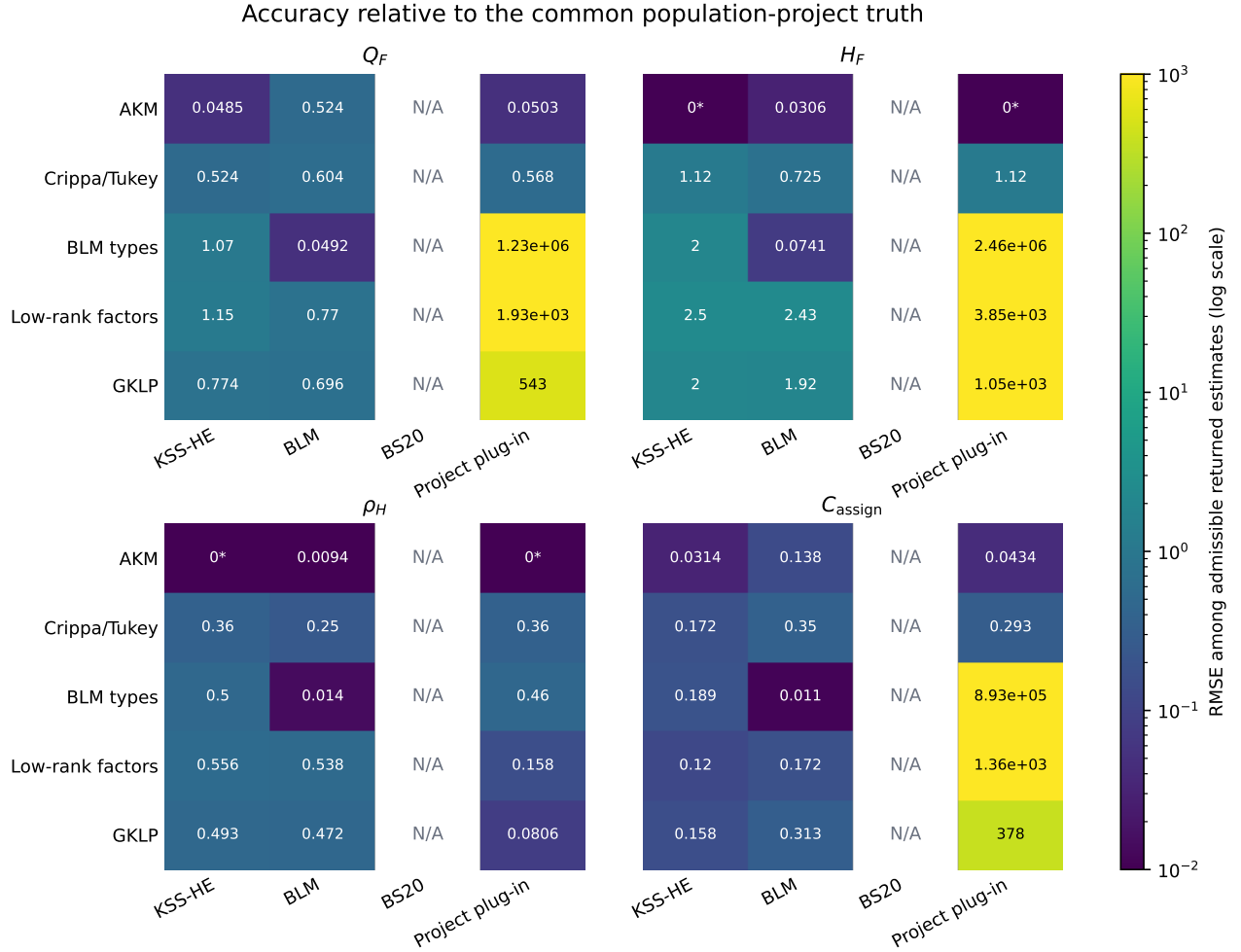


Figure 2: Main accuracy comparison: RMSE relative to the same population-project truth within each panel, among admissible returned estimates. Printed values give the RMSE; color uses a logarithmic scale so both ordinary and explosive errors remain visible. N/A means Table 3 gives no defensible mapping. The label 0* denotes an exact structural zero under the AKM DGP.

Three patterns matter. First, KSS’s H_F and ρ_H errors under the nonadditive DGPs are exactly the magnitudes excluded by additivity. Second, supported BLM fits recover the grouped DGP but lose substantial interaction variation when a continuous schedule is compressed into six cells. Its H_F RMSE is 0.725 under Crippa, 2.433 under continuous low rank, and 1.923 under GKLP. Third, the project estimator’s all-returned RMSE can be dominated by a small number of positive-rank warning fits. Figure 5 separates this tail risk from its pass-only behavior.

6.4 Native targets versus project targets

For procedure p , comparison with the project truth decomposes as

$$\hat{\theta}_p - \theta_{\text{proj}} = \underbrace{(\hat{\theta}_p - \theta_p)}_{\text{estimation error}} + \underbrace{(\theta_p - \theta_{\text{proj}})}_{\text{estimand difference}} . \quad (14)$$

Figure 3 and Table 7 report the second term averaged over all 100 populations. They contain no estimator sampling error. The table gives exact averages; the dot plot is included to make the sign and

relative size of the three kinds of target disagreement immediately visible. The three plotted contrasts are the direct native moments returned by KSS-HE and BS20. BLM’s four grouped-schedule gaps are discussed separately below because they measure discretization into evaluation cells rather than a native scalar named by BLM; the project plug-in has no gap because its target is the project truth by definition.

Table 7: Mean native population target minus project population target

DGP	KSS firm variance $-Q_F$	KSS covariance $-C_{\text{assign}}$	$C_{\text{BS}} - C_{\text{assign}}$
AKM	+0.000	+0.000	+0.416
Crippa/Tukey	−0.394	−0.0028	+0.824
BLM types	−0.752	+0.014	+0.622
Low-rank factors	−1.127	−0.017	+0.459
GKLP	−0.741	−0.080	+0.452

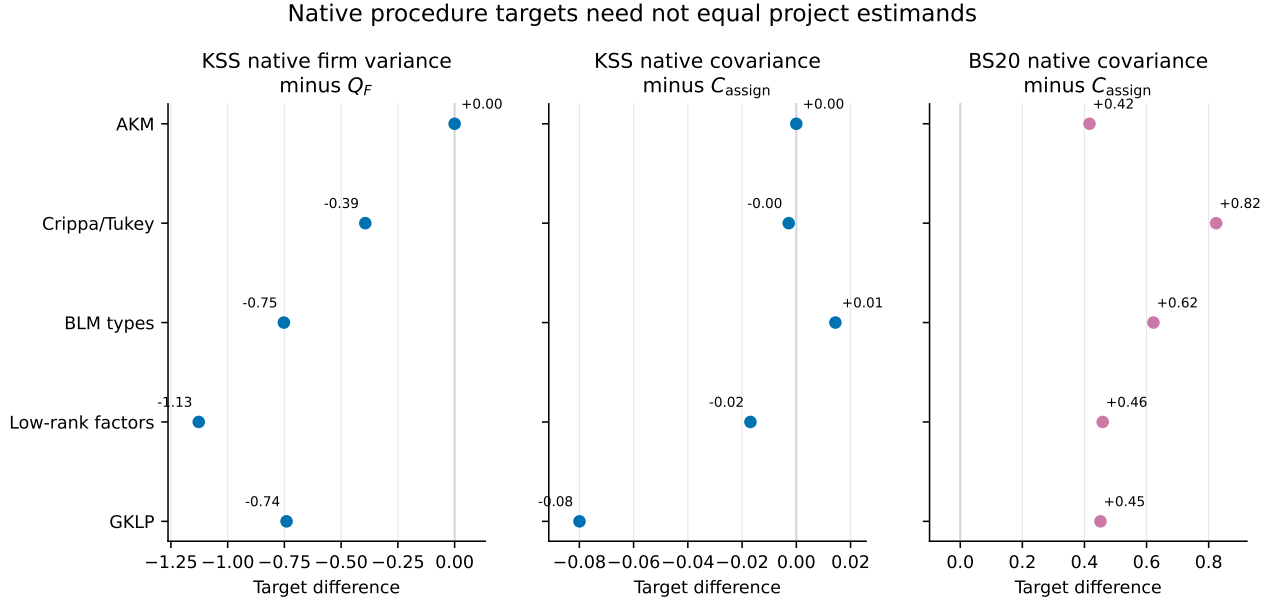


Figure 3: Population estimand differences, averaged over 100 generated populations. The vertical zero line marks agreement. These points are native targets minus project targets, not estimator bias.

The negative KSS firm-variance differences have an exact explanation under independent assignment. The additive projection then recovers b_j , so Equation (4) gives

$$\text{Var}_q(\psi_j) - Q_F = -\frac{H_F}{2} \leq 0. \quad (15)$$

With sorted assignment, the pseudo-true additive firm effect can absorb part of h_{ij} , so Equation (15) no longer fixes the sign. The negative values in the four sorted nonadditive designs are therefore informative simulation results, not a universal theorem. The KSS covariance gaps are small and of mixed sign, confirming that no analogous sign claim is warranted.

The BS20 object also needs its own explanation. Under an additive schedule $m_{ij} = a_i + b_j$, define the

conditional-expectation operators $Tb(i) = \mathbb{E}[b_J \mid I = i]$ and $T^*a(j) = \mathbb{E}[a_I \mid J = j]$. Then

$$\lambda_i = a_i + Tb(i), \quad \mu_j = b_j + T^*a(j), \quad (16)$$

$$C_{\text{BS}} - C_{\text{assign}} = \|Tb\|^2 + \|T^*a\|^2 + \langle Tb, TT^*a \rangle \geq 0. \quad (17)$$

Here T averages a firm quantity over the firms assigned to a worker, T^* averages a worker quantity over the workers assigned to a firm, $\|x\|^2$ is the relevant marginally weighted mean of x^2 , and $\langle x, z \rangle$ is the corresponding weighted cross-product. To see the sign, Cauchy–Schwarz and the mean-square contraction property of conditional expectation give $\langle Tb, TT^*a \rangle \geq -\|Tb\|\|T^*a\|$. The right side of Equation (17) is therefore at least $x^2 + y^2 - xy \geq 0$, with $x = \|Tb\|$ and $y = \|T^*a\|$. Thus the positive AKM gap has a mathematical explanation: sorting enters both conditional wage types. For a general nonadditive schedule, Equation (17) need not hold. The positive gaps in the other four rows are features of these DGPs, not a general property imposed by the code.

BLM has another population difference outside its discrete DGP. Replacing a continuous wage surface with a 2×3 grouped surface changes the project functionals even before estimation. The grouped-minus-full H_F gaps are -0.591 for Crippa, -2.452 for continuous low rank, and -1.959 for GKLP. The grouped BLM DGP has zero gap up to floating-point error because its true labels already define the scoring cells.

6.5 Rank selection

Rank is the number of interaction dimensions in Equation (9). Rank zero is additive; rank one permits one comparative-advantage dimension; rank two permits two. Figure 4 shows how often BIC selects each candidate. This quantity is displayed because a numerically converged fit can still omit a real interaction direction, creating structural approximation error before any LOO correction is considered. The table gives exact counts; the heatmap shows the selected-rank distribution. The orange outline marks the true rank, while the text at right is the separate multi-start diagnostic-pass rate.

Table 8: Project BIC rank selection

DGP	True rank	Select 0	Select 1	Select 2	Pass rate
AKM	0	100	0	0	100%
Crippa/Tukey	1	100	0	0	100%
BLM types	1	2	98	0	6%
Low-rank factors	2	0	100	0	78%
GKLP	1	0	100	0	94%

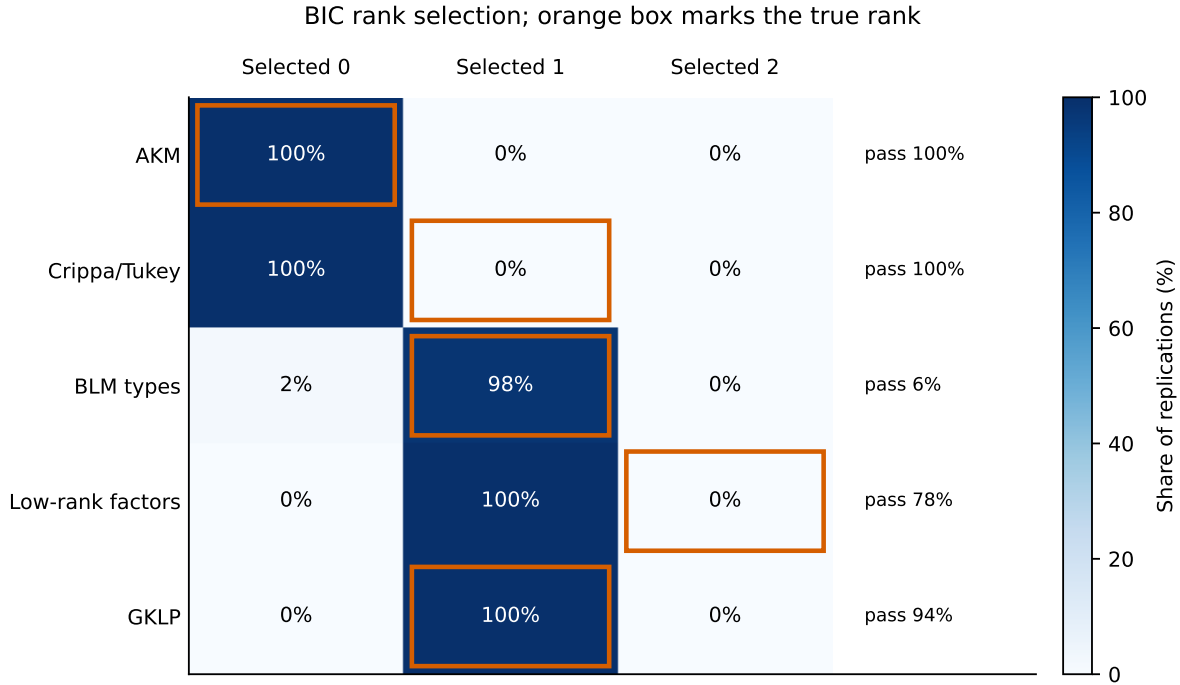


Figure 4: Share of 100 replications selecting each BIC rank. The orange outline marks the DGP’s true rank. The right-hand pass percentage concerns numerical agreement across starts, so it need not equal the rank-selection percentage.

Crippa makes the distinction sharp: every project fit passes numerically, but BIC selects rank zero in every replication. The estimator therefore sets the true rank-one interaction to zero. In the rank-two DGP, BIC always selects rank one, consistent with retaining the stronger singular-value-1 direction and discarding the weaker singular-value-0.5 direction. A future LOO bias correction cannot repair an interaction dimension removed by rank selection.

6.6 Warnings and tail risk

Figure 5 compares project RMSE among all returned fits with RMSE conditional on passing the numerical diagnostics. A large gap shows that a small warning subset dominates squared error. The pass-only line is useful for diagnosis, but it conditions on an estimation outcome and cannot replace all-returned performance. AKM is omitted because it has no project warnings; Crippa is retained as a reference case in which the two lines coincide. A logarithmic vertical axis is necessary because the warning fits make some RMSEs millions of times larger than the pass-only values.

Project plug-in: diagnostic warnings identify large tail risk

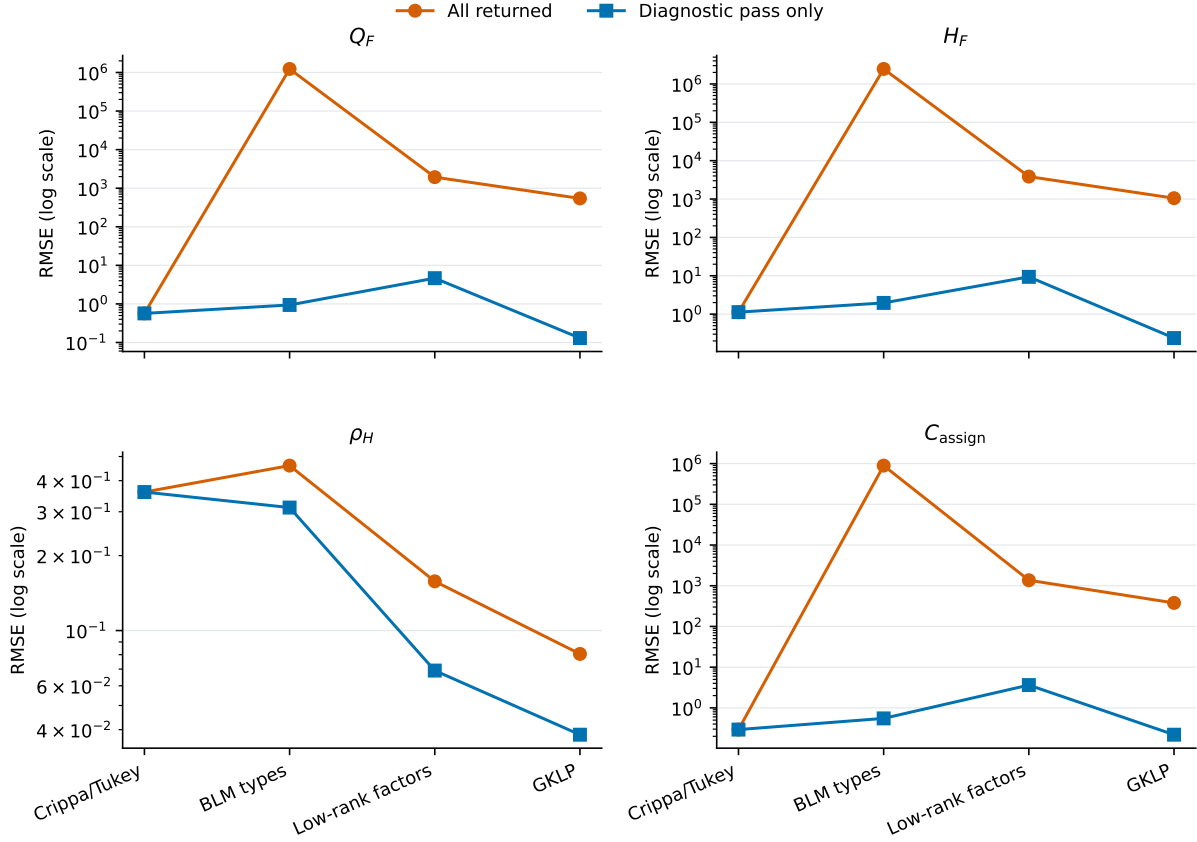


Figure 5: Project plug-in RMSE among all admissible returned fits (orange) and conditional on a diagnostic pass (blue). Each panel uses one project quantity; the vertical axis is logarithmic.

The lines coincide for Crippa because all 100 fits pass. In the grouped BLM DGP, by contrast, 94 warning fits raise the all-returned Q_F , H_F , and C_{assign} RMSEs from ordinary values below two to roughly 10^6 . The 22 low-rank warnings and six GKLP warnings likewise raise those three squared-wage metrics by several orders of magnitude. The much smaller gap for ρ_H does not make the underlying schedules stable; it reflects the normalization of interaction variation by total firm-side variation. Thus the warning label identifies economically consequential tail risk rather than a cosmetic convergence message.

7 Interpretation and next simulation changes

The full DGP-by-estimator matrix answers the original question more clearly than an all-AKM exercise. KSS works in its additive benchmark, and BLM works in its discrete benchmark when its support requirements are met. Outside those benchmarks, performance changes for two different reasons: estimators can be inaccurate for their native target, and native targets can differ from the project target. Equation (14) shows why those sources must not be combined without explanation.

The corrected BLM result is not “the optimizer fails half the time.” The current observation design produces too few stayers after no-return cleaning. More than half of the panels therefore do not identify every probability row that the configured 2×3 BLM model asks PyTwoWay to update. A fair next experiment should retain the present design as a support stress test and add a BLM-feasible design—for example, more workers, a permanent-stayer mixture, a lower redraw probability, or fewer firm classes. That is a DGP change and should be declared in advance rather than hidden inside the estimator.

For the project procedure, the dominant present issue is rank selection and positive-rank reproducibility, not only small-sample plug-in bias. Crippa shows complete under-selection with numerical convergence; the rank-two design shows systematic truncation; and the warning fits create very large squared-error tails. Rank choice and stable schedule completion therefore need to be solved before a leave-out correction can be interpreted as the principal remaining improvement.

Finally, the reporting distinctions are substantive. KSS’s interaction zeros are now labeled “imposed,” BS20 is reported under its native covariance rather than under C_{assign} , and BLM support failures are separate from optimizer failures. These labels prevent a numerical table from silently changing the meaning of an estimand.

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